

EXAMINING THE EXTERNAL TRAINING LOAD OF AN ENGLISH PREMIER LEAGUE FOOTBALL TEAM WITH SPECIAL REFERENCE TO ACCELERATION

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ABSTRACT

Akenhead, R, Harley, J, and Tweddle, S. Examining the external training load of an English Premier League football team with special reference to acceleration. *J Strength Cond Res* 30(9): 2424–2432, 2016—Practitioners and coaches often use external training load variables such as distance run and the number of high-speed running (HSR) activities to quantify football training. However, an important component of the external load may be overlooked when acceleration activities are not considered. The aim of this study was to describe the within-microcycle distribution of external load, including acceleration, during in-season 1-game weeks in an elite football team. Global Positioning System technology was used to collect time-motion data from 12 representative 7-day microcycles across a competitive season (48 training days, 295 data sets). Training time, total distance (TD), high-speed running (HSR) distance ($>5.8 \text{ m} \cdot \text{s}^{-1}$), sprint running distance ($>6.7 \text{ m} \cdot \text{s}^{-1}$) and acceleration variables were recorded during each training session. Data were analysed for interday and interposition differences using mixed linear modeling. The distribution of external load was characterized by the second training day of the microcycle (5 days prematch) exhibiting the highest values for all variables of training load, with the fourth day (1 day prematch) exhibiting the lowest values. Central midfield players covered $\sim 8\text{--}16\%$ greater TD than other positions excluding wide midfielders ($p \leq 0.03$, $d = 0.2\text{--}0.4$) and covered $\sim 17\%$ greater distance accelerating $1\text{--}2 \text{ m} \cdot \text{s}^{-2}$ than central defenders ($p = 0.03$, $d = 0.7$). When expressed relative to training duration and TD, the magnitude of interday and interposition differences were markedly reduced ($p = 0.03$, $d = 0.2\text{--}0.3$). When managing the distribution of training load, practitioners should be

aware of the intensity of training sessions and consider the density of external load within sessions.

KEY WORDS acceleration, soccer, time motion analysis, training

INTRODUCTION

The objective of the training process in football is to administer an appropriate frequency, volume, and intensity of training to deliver the appropriate psychological and physiological stimuli to achieve adaptations which will improve individual and team performance. With regard to the physiological component, the stimuli for adaptations are determined by the relative physiological load, often termed the internal training load (ITL) (25). Within team sports, ITL is commonly quantified using heart rate indices (39) or ratings of perceived exertion (RPE) (20). Although heart rate indices may reflect the aerobic load, when considered alone, it is not an appropriate measure of total load for football because of the intermittent activity profile and the substantial contribution from anaerobic energy sources (9). The reliability of RPE has also been questioned and remains to be validated during football competition which may explain the recently reported low utilization of RPE within high-level senior football (5,36,39). Consequently, practitioners often quantify the frequency, volume, and density of the external training load (ETL) components such as total distance (TD) covered or number of sprints executed to provide an objective and more comprehensive understanding of football training.

The prescription and distribution of ETL within professional football is heavily influenced by competition frequency, with in-season microcycles of typically 3–7 days in duration. Therefore, following the preseason period in which fitness capacities are reestablished, a “multiple peaking” periodization model is required in order for players to repeatedly perform at a high level in all matches across the season (31,33). When planning a training program, coaches must not only consider the technical and tactical objectives but also the positive (improved fitness) and negative (increased fatigue) consequences of each individual training

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TABLE 1. Time-motion, heart rate, and accelerometer parameters describing the training weeks examined.*†

	MD – 5 (<i>n</i> = 68)	MD – 4 (<i>n</i> = 79)	MD – 3 (rest)	MD – 2 (<i>n</i> = 77)	MD – 1 (<i>n</i> = 71)	Match	Total (<i>n</i> = 295)
Duration (min)‡	76 ± 16 (72–80)	87 ± 8 (83–91)		81 ± 12 (77–85)	60 ± 11 (56–64)		304 (288–320)
TD (m)‡	4,925 ± 1,312 (4,650–5,200)	6,234 ± 1,368 (5,966–6,501)		5,287 ± 1,363 (5,017–5,556)	3,493 ± 1,324 (3,221–3,765)		19,939 (18,854–21,022)
Walk/jog (m)‡	3,691 ± 989 (3,484–3,900)	4,568 ± 1,030 (4,365–4,770)		3,975 ± 1,027 (3,771–4,179)	2,829 ± 1,000 (2,623–3,035)		15,063 (14,243–15,884)
HSR distance (m)	44 ± 160 (11–77)	190 ± 165 ^{a,c,d} (158–222)		141 ± 165 ^{a,d} (109–174)	23 ± 162 (0–56)		398 (278–529)
HSR efforts (<i>n</i>)	5 ± 9 (3–7)	14 ± 10 ^{a,c,d} (13–16)		11 ± 10 ^{a,d} (9–13)	3 ± 10 (1–5)		33 (26–41)
SPR distance (m)	7 ± 50 (0–17)	46 ± 51 ^{a,c,d} (36–55)		28 ± 51 ^{a,d} (18–37)	6 ± 50 (0–16)		87 (54–125)
SPR efforts (<i>n</i>)	1 ± 3 (0–1)	3 ± 3 ^{a,c,d} (3–4)		2 ± 3 ^{a,d} (1–3)	0 ± 3 (0–1)		6 (4–9)
ACC _{TOTAL} (m)	473 ± 134 (446–501)	574 ± 138 ^{a,c,d} (547–600)		469 ± 138 ^d (442–495)	310 ± 134 (282–337)		1,826 (1,717–1,933)
DEC _{TOTAL} (m)	417 ± 92 (393–442)	501 ± 87 ^{a,c,d} (477–525)		411 ± 78 ^d (387–435)	269 ± 72 (245–293)		1,598 (1,502–1,695)
>90%HRmax (min)	7 ± 7 ^{c,d} (5–9)	7 ± 8 ^d (5–9)		4 ± 5 (2–6)	3 ± 5 (1–5)		21 (13–29)
Player load (a.u.)	536 ± 161 (503–569)	622 ± 168 ^{a,c,d} (590–654)		543 ± 167 ^d (511–576)	392 ± 162 (360–425)		2,093 (1,964–2,224)

*MD – *x* = *x* days before the match; TD = total distance; Walk/jog distance = <3.0 m·s⁻¹; HSR = high speed running >5.8 m·s⁻¹; SPR = sprint running >6.7 m·s⁻¹; ACC_{TOTAL} and DEC_{TOTAL} are distances covered >1 m·s⁻² and <–1 m·s⁻² respectively; HR = heart rate; a.u. = arbitrary units.

†Data are presented as mean ± SD (95% confidence interval).

‡All days different from one another; a, greater than MD – 5; c, greater than MD – 2; d, greater than MD – 1 (*p* ≤ 0.05).

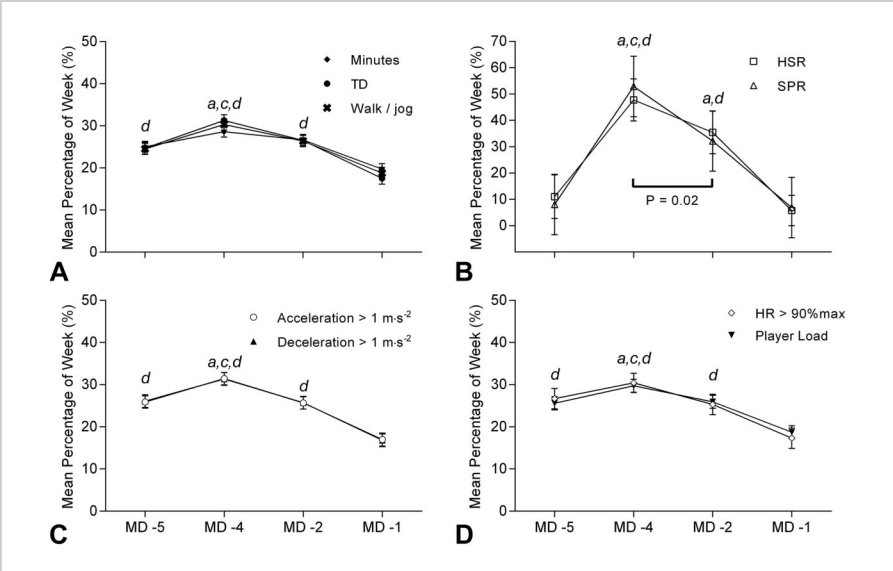


Figure 1. Mean percentage distribution of measured variables across representative training weeks. Error bars represent 95% confidence intervals. a, greater than MD - 5; c, greater than MD - 2; d, greater than MD - 1. All differences are $p < 0.001$ unless otherwise stated. TD, total distance; HSR, high-speed running distance; SPR, sprint running distance; MD - x , x days before the match.

session and the net result of the accumulated load on match day. For instance, higher volume and intensity training tends to be prescribed early in the microcycle, with training volume decreasing as match day approaches to facilitate decay of the fatigue component (25,27). Opportunities to measure

utilized GPS to describe ETL across 29 training sessions for a team of Australian professional football players. The authors reported that players covered $4,467 \pm 1,296$ m TD with 132 ± 101 m accumulated at a speed $>5.5 \text{ m} \cdot \text{s}^{-1}$ on average per session. However, the analysis did not include

the physiological or performance response of players to a given ETL dose can be limited within the applied environment, and so practitioners often try to uphold the principles of tapering training load (reduction in volume with a maintenance or small reduction in intensity) through manipulation and evaluation of ETL (5).

Widespread use of micro-technology such as the Global Positioning System (GPS) and other microelectromechanical system (MEMS) devices has given practitioners access to more objective ETL data than ever before (16). In light of the abundance of available data, a challenge lies in identifying those variables which provide useful information about training load. Scott et al. (35) utilized

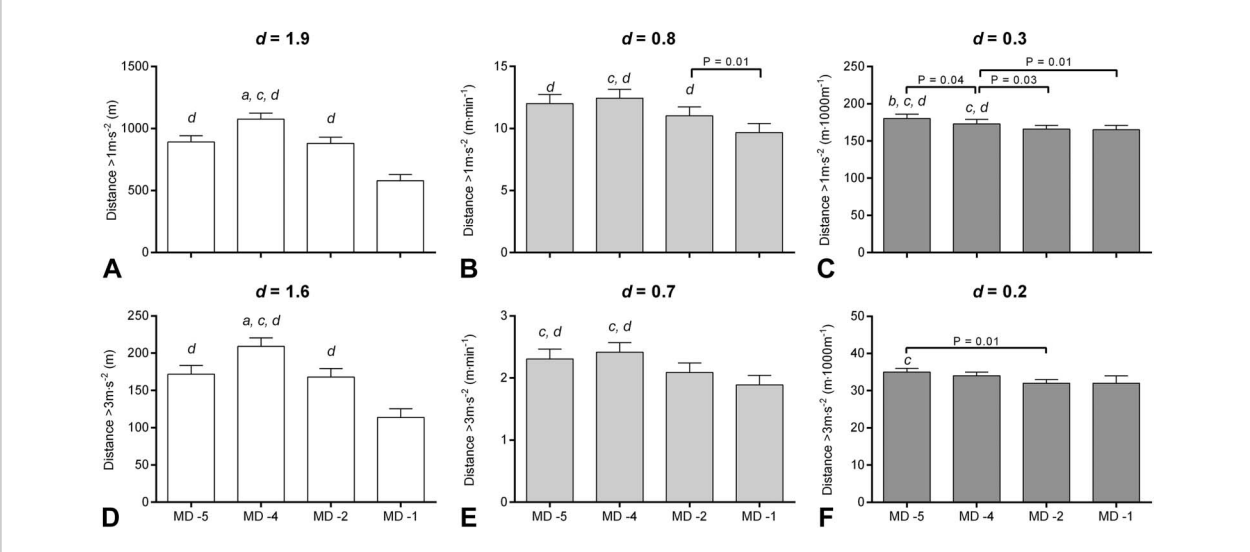


Figure 2. Combined acceleration and deceleration distance ($>1 \text{ m} \cdot \text{s}^{-2}$) expressed in absolute values (A); per minute (B); per 1,000 m of total distance covered (C). Combined acceleration and deceleration distance ($>3 \text{ m} \cdot \text{s}^{-2}$) expressed in absolute values (D); per minute (E); per 1,000 m of total distance covered (F). a, greater than MD - 5; b, greater than MD - 4; c, greater than MD - 2; d, greater than MD - 1. All differences are $p < 0.001$ unless otherwise stated. MD - x , x days before the match.

acceleration activities and did not examine the distribution of ETL within microcycles. Similarly, Gaudino et al. (21) reported mean positional differences in TD and metabolic power variables in an elite team but did not quantify acceleration activities or describe the distribution of ETL within microcycles. More recently, Malone et al. (27) reported ETL (TD, average speed, high-speed distance [$>5.5 \text{ m} \cdot \text{s}^{-1}$]) and ITL data (RPE, average percentage of maximum heart rate [%HRmax]) from an elite team across a full season. The final training session of the microcycle (1 day pre-match) featured the lowest values for all ETL and ITL variables with the exception of high-speed distance. Interposition differences were found for TD and high-speed distance, but acceleration activities were not included in the analysis.

Existing studies have described the TDs or distances covered above predefined speed thresholds within training sessions and microcycles for individual teams; however, none have included information regarding acceleration activities. Additionally, the rate of accumulation of load (e.g., per minute) is an important determinant of the resultant intensity of the exercise. For instance, accelerations occur frequently in football (4,23,37) and a greater density of accelerations is associated with a higher physiological and neuromuscular demand which is not considered when quantifying only absolute values or speed-based variables. Therefore, the aims of this study were to (a) describe the distribution of ETL during in-season microcycles of an elite football team with a particular focus on acceleration and (b) examine the interday and interposition variation in ETL variables within representative microcycles.

METHODS

Experimental Approach to the Problem

Permission was obtained from a professional football club competing in the English Premier League to carry out analysis on training load data collected during training sessions of the 2011–2012 season in which the team finished fifth. The competitive season lasted 39 weeks, of which 27 weeks (69%) contained a single game, 8 weeks featured 2 games, and 4 weeks featured 0 games. Of the 27 single-game weeks, 18 (67%) were characterized by a 7-day microcycle and therefore represented the most prevalent microcycle within the season. To facilitate standardization within the analysis, the following eligibility criteria were imposed: (a) one competitive game within the week, (b) 4 training sessions within the week, and (c) a day of rest followed both the game and the second training session of the week. The inclusion criteria identified 12 microcycles which were evenly distributed throughout the competitive season (from week 10 to week 39). The training sessions of the selected microcycles were categorized using the “match day minus” format (27); for instance, match day minus 1

TABLE 2. Mean daily time-motion, heart rate, and accelerometer parameters for each playing position examined.*†

	CD	WD	CM	WM	F
TD (m)	4,834 (4,444–5,224)	4,846 (4,488–5,204)	5,463‡ (5,093–5,833)	4,971 (4,607–5,337)	4,809 (4,446–5,171)
Walk/jog (m)	3,919 (3,650–4,190)	3,600 (3,346–3,854)	3,931 (3,670–4,194)	3,680 (3,418–3,942)	3,698 (3,439–3,957)
HSR distance (m)	79 (37–121)	118 (80–158)	102 (62–144)	102 (61–144)	94 (54–135)
HSR efforts (n)	6 (3–10)	10 (7–13)	8 (4–11)	9 (6–12)	8 (5–11)
SPR distance (m)	19 (4–35)	28 (14–42)	13 (0–28)	22 (9–37)	24 (10–38)
SPR efforts (n)	1 (1–2)	2 (1–3)	1 (0–2)	2 (1–2)	2 (1–3)
ACC _{TOTAL} (m)	421 (372–469)	463 (422–504)	502‡ (459–545)	461 (419–502)	435 (394–477)
DEC _{TOTAL} (m)	360 (319–403)	405 (369–442)	441 (404–480)	411 (374–448)	378 (342–416)
>90%HRmax (min)	5.1 (1.3–8.9)	4.2 (1.1–7.4)	7.6 (3.1–11.0)	5.8 (2.5–9.2)	3.4 (1.0–6.7)
Player load (a.u.)	511 (440–580)	495 (440–550)	582 (520–640)	513 (455–570)	517 (460–575)

*CD = central defenders; WD = wide defenders; CM = central midfielders; WM = wide midfielders; F = forwards; TD = total distance; Walk/jog distance = $<3.0 \text{ m} \cdot \text{s}^{-1}$; HSR = high-speed running $>5.8 \text{ m} \cdot \text{s}^{-1}$; SPR = Sprint running $>6.7 \text{ m} \cdot \text{s}^{-1}$; ACC_{TOTAL} and DEC_{TOTAL} are distances covered $>1 \text{ m} \cdot \text{s}^{-2}$ and $<-1 \text{ m} \cdot \text{s}^{-2}$ respectively; HR = heart rate; a.u. = arbitrary units.

†Data are presented as mean (95% confidence interval).

‡Greater than CD, WD and F ($p < 0.03$).

refers to the training session 1 day before the match (MD - 1).

Subjects

Inclusion criteria for individual data sets specified that players must be fit and available for team selection (i.e., not in the return to play process) and must have completed the full training session. These criteria identified 33 players (24 ± 4 years; 82 ± 8 kg; 1.83 ± 0.05 m), yielding 295 data sets with each player included between 1 and 21 times (central defender [CD] = 52, wide defender [WD] = 73, central midfielder [CM] = 56, wide midfielder [WM] = 54, forward [F] = 60 data points). The study was approved by the institutional ethics committee at Northumbria University, Newcastle upon Tyne, United Kingdom.

Procedures

Values for TD, high-speed running (HSR) distance (≥ 5.8 m·s⁻¹), sprint running distance (SPR ≥ 6.7 m·s⁻¹), and walk/jog distance (< 3.0 m·s⁻¹) were extracted using the manufacturer's supplied software (Sprint 5.1; Catapult, Melbourne, Australia). The number of HSR and SPR efforts were also recorded. In accordance with Osgnach et al. (32) and Minetti et al. (30), the distance covered accelerating and decelerating was analysed using operationally defined thresholds of 1–2 m·s⁻² (low), 2–3 m·s⁻² (moderate), and > 3 m·s⁻² (high). Total acceleration (ACC_{TOTAL}) and total deceleration (DEC_{TOTAL}) were classified as distances covered > 1 m·s⁻² and < -1 m·s⁻², respectively. Acceleration and deceleration were also pooled at $> \pm 1$ m·s⁻² ($> 1_{TOTAL}$) and $> \pm 3$ m·s⁻² ($> 3_{TOTAL}$) to provide a measure of total velocity change load. The PlayerLoad variable, calculated as the sum of

vertical, lateral, and anteroposterior forces using the integrated 100-Hz triaxial accelerometer housed within the GPS unit was also recorded (10).

Rating of perceived exertion was not utilized by the team at the time of data collection. Time spent over 90% of maximum heart rate ($> 90\%HR_{max}$) was recorded and is included in the analysis (Team 2; Polar Electro Oy, Finland). The players' individual maximum heart rate used for this calculation was taken from preseason maximal testing to volitional exhaustion using the Yo-Yo Intermittent Recovery Test Level 2 (8). It is acknowledged that time $> 90\%HR_{max}$ is not strictly a measure of ITL; however, there exists a rationale for its use based on the dose-response relationship observed between the time spent above this threshold and improvements in fitness (14).

Time-motion data were recorded at 10 Hz for the entire training session including the warm-up period as previously described (Catapult MiniMaxx S4, Firmware 6.7; Catapult, Australia) (4). For the calculation of acceleration, a smoothing filter of 0.5 s was applied to GPS data using the manufacturer's software. The number of satellites connected to devices was 12 ± 1 , with a horizontal dilution of precision of 0.9 ± 0.1 . The validity and reliability of the MEMS devices used have been established in the literature (2,38) and are acceptable for the current variables and thresholds used (calculation of acceleration; standard error of the estimate = $0.12\text{--}0.19$ m·s⁻²; typical error as a coefficient of variation % [TE%] = $3.9 \pm 0.4\%$).

Statistical Analyses

Variables were expressed in absolute values and relatively with respect to the training time (per minute) and distance (per 1,000 m of TD in session). All data satisfied the criteria for normal distribution (D'Agostino-Pearson Omnibus K2 test) with the exception of HSR distance and efforts, and SPR distance and efforts which contained a disproportionate number of "natural" zero values. However, justification exists to include these zero values as they partially define the ETL of the day in which they occurred.

Mixed linear modeling with a Sidak adjustment was conducted to examine interday, interposition, and day-position interactions in each variable using IBM SPSS for Windows v.18.0 (significance accepted at $p \leq 0.05$). The effect size (Cohen's d) and 95%

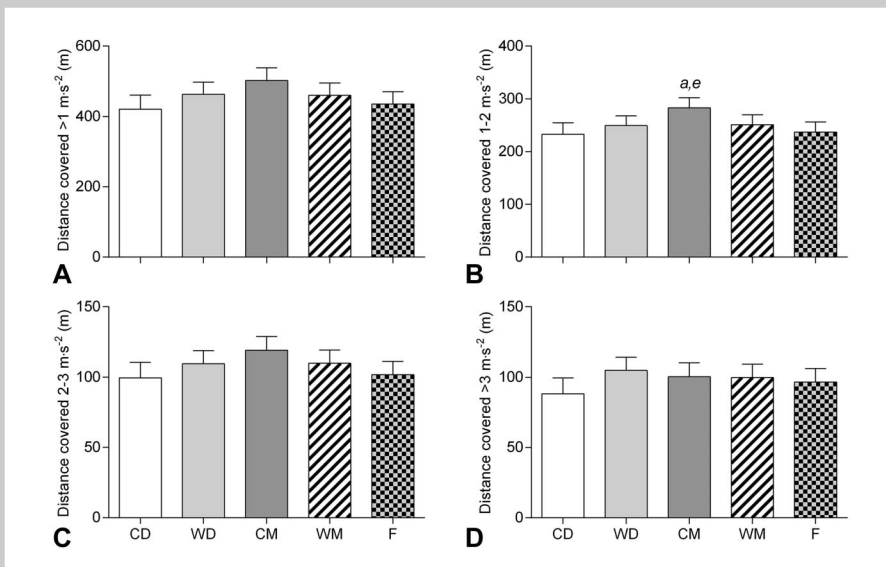


Figure 3. Mean distance $\pm$ 95% confidence interval covered accelerating > 1 m·s⁻² (ACC_{TOTAL}) (A); 1–2 m·s⁻² (low) (B); 2–3 m·s⁻² (moderate) (C); > 3 m·s⁻² (high) (D); a, greater than central defender (CD); e, greater than forward (F) ($p \leq 0.05$). WD, wide defender; CM, central midfielder; WM, wide midfielder.

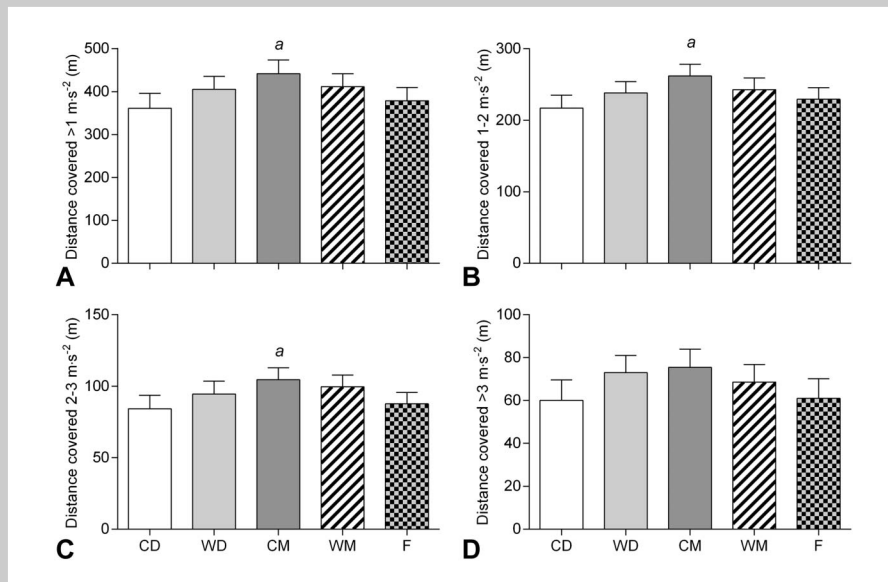


Figure 4. Mean distance $\pm$ 95% confidence interval covered decelerating $> 1 \text{ m} \cdot \text{s}^{-2}$ ($\text{DEC}_{\text{TOTAL}}$) (A); $1-2 \text{ m} \cdot \text{s}^{-2}$ (low) (B); $2-3 \text{ m} \cdot \text{s}^{-2}$ (moderate) (C); and $> 3 \text{ m} \cdot \text{s}^{-2}$ (high) (D). a, greater than central defender (CD) ($p \leq 0.05$). WD, wide defender; CM, central midfielder; WM, wide midfielder; F, forward.

confidence interval (CI) were also calculated for interday and interposition differences. The effect size was defined as trivial (< 0.2), small ($0.2-0.5$), moderate ($0.5-0.8$), and large (> 0.8) (15). The Pearson product moment correlation coefficients (r) were also calculated to provide a measure of

association between the ETL variables. Intraday and intraposition variation for all variables was established by calculating the TE (as a coefficient of variation %) of variables from the data of players completing at least 3 full weeks of training ($n = 15$). Values are reported as mean (95% CI) unless otherwise stated.

RESULTS

Analysis revealed interday differences in several variables ($p \leq 0.05$); however, there were no interactions between day and playing positions ($p = 0.35$). Intraday interplayer TE was 9–25% for TD, 52–156% for HSR, 82–246% for SPR, and 13–40% for all acceleration variables. Intraday intraposition TE for the acceleration and deceleration zones used was 20% for low, 23% for moderate, and 25% for high. Typical error was greater for the number of acceleration efforts compared with distance covered in the moderate ($p = 0.02$, $d = 0.7$) and high ($p = 0.001$, $d = 1.4$) zones. Intraday intraposition TE did not differ across days although MD – 1 tended to exhibit the greatest TE ($p = 0.39$, $d = 0.6$).

Interday Analysis

Main interday effects were present for all variables when expressed in absolute terms (Table 1) and as a percentage of the total weekly load (Figure 1, $F_{3,260} = 23-210$, $p = 0.001$). Post-hoc analysis revealed the highest values occurred on MD – 4, with the lowest values on MD – 1 (Figure 1). Figure 2 shows the between-day differences in $> 1_{\text{TOTAL}}$ and $> 3_{\text{TOTAL}}$ when expressed in absolute and relative terms.

Positional Analysis

Main interposition effects were present for TD; total, low, and moderate deceleration distances; and low acceleration distance ($F_{4,20} = 3.2-4.9$, $p \leq 0.05$,

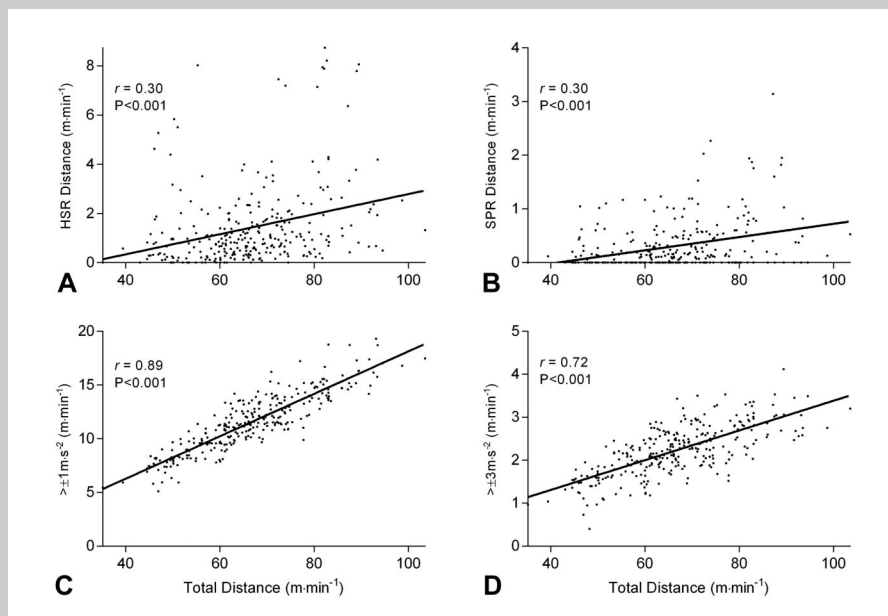


Figure 5. The association between total distance and high-speed running (HSR) distance ($> 5.8 \text{ m} \cdot \text{s}^{-1}$) (A); sprint running (SPR) distance ($> 6.7 \text{ m} \cdot \text{s}^{-1}$) (B); $> 1_{\text{TOTAL}}$ (accelerations and decelerations $\geq \pm 1 \text{ m} \cdot \text{s}^{-2}$) (C) and $> 3_{\text{TOTAL}}$ (accelerations and decelerations $\geq \pm 3 \text{ m} \cdot \text{s}^{-2}$) (D). r = Pearson moment correlation coefficient.

Table 2). Figures 3 and 4 illustrate the positional differences in acceleration and deceleration distance variables, respectively. Central midfield players covered a greater distance within the low acceleration threshold compared with CD and F (mean difference 47 m; 95% CI 3–97 m, $d = 0.5$, $p = 0.03$). Central midfield players also covered a greater total, low, and moderate deceleration distances compared with CD with mean differences of 81 m (95% CI 5–157 m, $d = 0.5$, $p = 0.03$), 45 m (95% CI 7–83 m, $d = 0.4$, $p = 0.01$), and 21 m (95% CI 0–41 m, $d = 0.6$, $p = 0.04$). When expressed relative to TD, interposition differences in acceleration and deceleration variables were not significant ($p > 0.14$), but WD and WM players exhibited 10–20% greater acceleration density than CM and CD for high acceleration and deceleration ($d = 0.4$ – 0.6). Interposition differences for HSR distance and SPR distance were trivial to small ($p > 0.30$, $d = 0.1$ – 0.3).

Associations Between Variables

Figure 5 shows the association between average speed ($\text{m} \cdot \text{min}^{-1}$) and the rate of accumulation for HSR, SPD, $>1_{\text{TOTAL}}$ (pooled acceleration and deceleration $>1 \text{ m} \cdot \text{s}^{-2}$), and $>3_{\text{TOTAL}}$ (pooled acceleration and deceleration $>3 \text{ m} \cdot \text{s}^{-2}$).

DISCUSSION

The main findings of the current study were (a) only total distance, acceleration, and deceleration distances were able to differentiate between playing positions and (b) expressing acceleration and deceleration variables relative to training time and TD reduced the effect size of interday differences and altered the rank-order of training sessions within the microcycle.

Owing to differences in reported variables, comparison between current findings and previous studies is limited to TD. The mean TD in the current study is ~ 7 – 20% ($d = 0.2$ – 0.7) less than values reported by Malone et al. (27) for microcycles of the seventh, 24th, and 39th weeks of the season in a Premier League team. Conversely, the average session TD values are $\sim 13\%$ greater than the $4,467 \pm 1,296 \text{ m}$ reported by Scott et al. (35). Despite differences in absolute values between studies, the same trend for microcycle structure is present. In the current study, the second day of training (MD – 4) consistently produced the greatest absolute values in all variables, with the final training session of the week producing the lowest values (MD – 1). These findings are in general agreement with previous studies which have reported higher training loads at the beginning of the microcycle followed by a reduction in ETL before the match (25–27). Notwithstanding the different microcycle structures used, this seems to be a common strategy to “unload” players leading up to the match in an attempt to facilitate the decay of accumulated fatigue and promote readiness to perform.

The same periodization trend was also evident in acceleration and deceleration values. Absolute interday differ-

ences were large (46% reduction from MD – 4 to MD – 1; $d = 1.6$ – 1.9) but diminished when expressed relative to training duration and total training distance, indicating a reduced volume but maintenance of density. Despite smaller effects when expressed relatively (10% reduction from MD – 4 to MD – 1; $d = 0.2$ – 0.3), significant differences were still present between days, suggesting that training duration and total distance largely, but not exclusively, moderate these variables. This is supported by the high but not perfect association between these variables ($r = 0.72$ – 0.89). Few studies have directly investigated the effect of acceleration density, but Akenhead et al. (3) demonstrated that an increased acceleration density elevates physiological and perceived exertion in trained football players. Other studies have investigated the effects of changing direction on $\dot{V}\text{O}_2$, muscle oxygenation, blood lactate concentration, and perceived exertion, reporting that frequently changing direction and the inherent deceleration and acceleration involved substantially elevate the anaerobic contribution and overall physiological stress (6,13,22).

The absence of interposition differences in HSR and SPR variables observed in the current analysis does not reflect the well-established positional differences reported during match play. Sprint distance has been shown to differentiate between most positions during match play (18,19), with CD typically accumulating the lowest values compared with other positions (approximately 30–50% less HSR and SPR distances). The current data indicate intraweek interposition variation in SPR approximated only 32% ($d = 0.4$) compared with a positional variation of approximately 200% ($d = 1.2$) reported during matches in previous research (11,12). However, the current study is consistent with the findings of Gaudino et al. (21) who reported that CM players covered more TD than all other positions, but distance covered $>7 \text{ m} \cdot \text{s}^{-1}$ did not differ between any positions during preseason training in Premier League players. Similarly, Malone et al. (27) reported only small interposition differences in distance covered $>5.5 \text{ m} \cdot \text{s}^{-1}$, suggesting the current findings may be typical of ETL distribution in elite teams.

Despite no clear interposition differences in HSR and SPR, some acceleration variables were able to differentiate between positions. Central midfield players covered more distance within total, low, and moderate acceleration thresholds than CD (Figures 3 and 4). However, when expressed relative to TD, it was found that WD exhibited a greater density of accelerations and decelerations across all zones than CM ($+10$ – 20% , $d = 0.4$ – 0.6). This knowledge of how distance is accumulated is important to the practitioner and may be used to inform conditioning practices (e.g., the range of frequencies and magnitudes of discrete movements). For instance, a high utilization of small-sided games (SSGs) with limited variation of pitch dimensions, rules, or player numbers may partially contribute to the relatively homogeneous ETL faced by players of different positions. Dellal et al. (17) reported that 4v4 SSG lead to interposition differences in

TD, HSR, and SPR of only 10%, which was reduced to only 5% when CDs were excluded. Unpublished match data from our group suggest positional differences in acceleration distance approximates 10–20% for $>1 \text{ m} \cdot \text{s}^{-2}$ ($\text{ACC}_{\text{TOTAL}}$) and 25–40% for $>3 \text{ m} \cdot \text{s}^{-2}$ ($>3_{\text{TOTAL}}$). In the current study, the interposition variation of $\text{ACC}_{\text{TOTAL}}$ was similar to match values (12%, $d = 0.4$) but $>3_{\text{TOTAL}}$ was only 6% ($d = 0.2$). These data underscore the importance of considering individual player training profiles relative to their match profile for acceleration as well as HSR variables.

Supplementary individualized training may therefore be required to provide overload of movement frequencies and intensities not addressed by general squad training. This can also include resistance training which has proven effective in eliciting physiological adaptations which can enhance intermittent multidirectional running performance in football players (29,34). However, coaches should not simply “chase the numbers” when trying to achieve targets for ETL components. The on-field stimuli should be delivered in position-specific circumstances where possible to enhance transfer to match performance. For instance, training exercises that deliver the desired frequency and magnitude of accelerations but which also require players to anticipate and react to opponents are preferred over change of direction drills such as shuttle running for this purpose. In this respect, the practitioner can guide the coach on suitable training prescription by compiling ETL profiles for individual training exercises, taking into consideration the associated acceleration density and ITL.

The ITL of football training is difficult to quantify because of a host of reasons. The team studied used 90%HRmax as a proxy of ITL due to its relationship with improvements in football-specific fitness. The reported duration of ~21 minutes spent $>90\%$ HRmax represents ~7% of total weekly training time. This value may intuitively seem very low but is comparable to the percentage time spent within this threshold reported for youth (40) and senior professional players (7,28). The data of Wrigley et al. (40) included match exposure, and although the percentage time reported is similar to the current study, the absolute time $>90\%$ HRmax approximated 45 minutes per week. Castagna et al. (14) advised that players spend at least 8% of total training time $>90\%$ HRmax but did not provide recommendations for the absolute time spent within this zone or the distribution of this load within a microcycle.

The current analysis has caveats which must be considered when interpreting the findings. The current data are reflective of the paradigms and practices of the club and the training status of the players examined. Consequently, generalization to other teams or populations is not recommended. In addition, the use of arbitrary speed and acceleration thresholds restricts interpretation of relative physiological demands (1,24). There is currently no consensus on how to appropriately establish individual limits from which to express relative intensities for acceleration

density such as those observed during linear running (1,32). Nevertheless the data presented here add to the growing body of applied research and provide an alternative perspective when planning and analyzing the distribution of training load within elite football.

PRACTICAL APPLICATIONS

The current study supports the argument that the monitoring of only speed-based locomotor variables may not be sufficient to understand the complexity of elite football training. The quantification of acceleration variables within team sports training is warranted and provides the practitioner with important additional information regarding the prescription and distribution of training. Also worthy of consideration is the way in which variables are expressed, as the density of activities may provide more pertinent information when analyzing ETL and designing training programs.

The observed interposition differences in external load variables were smaller than those frequently reported within the literature for competitive matches. It is recommended that the individual player profiles created by practitioners include acceleration and deceleration output from training and match data where possible.

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